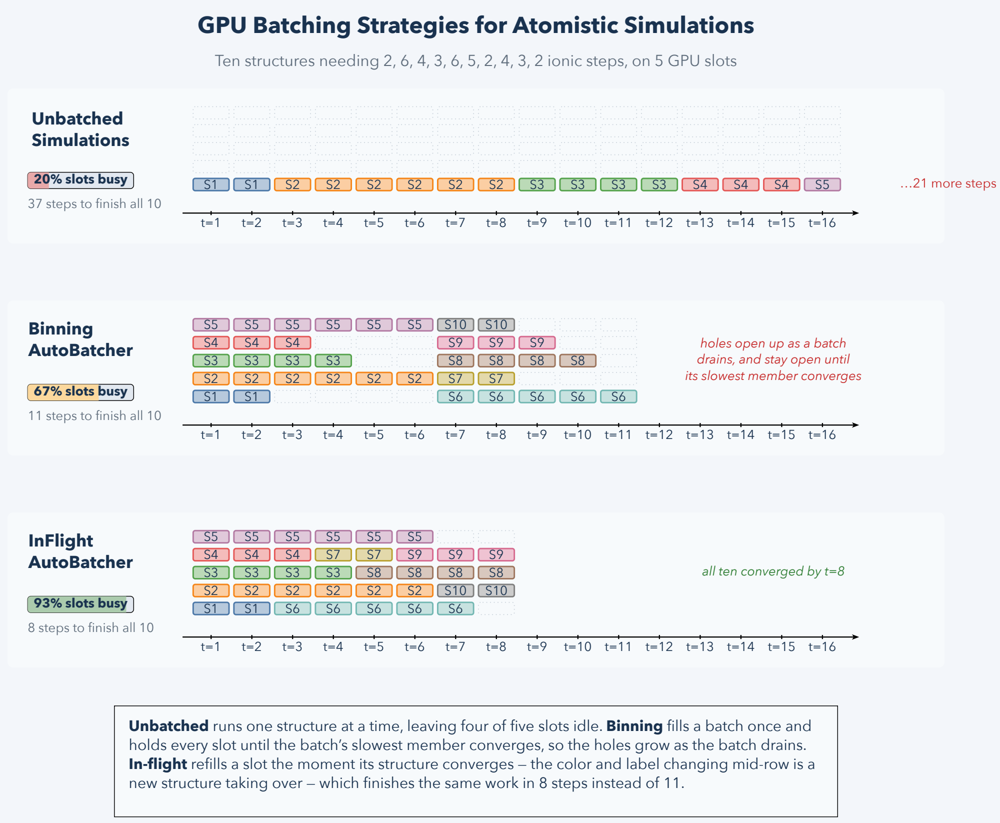
GPU Batching

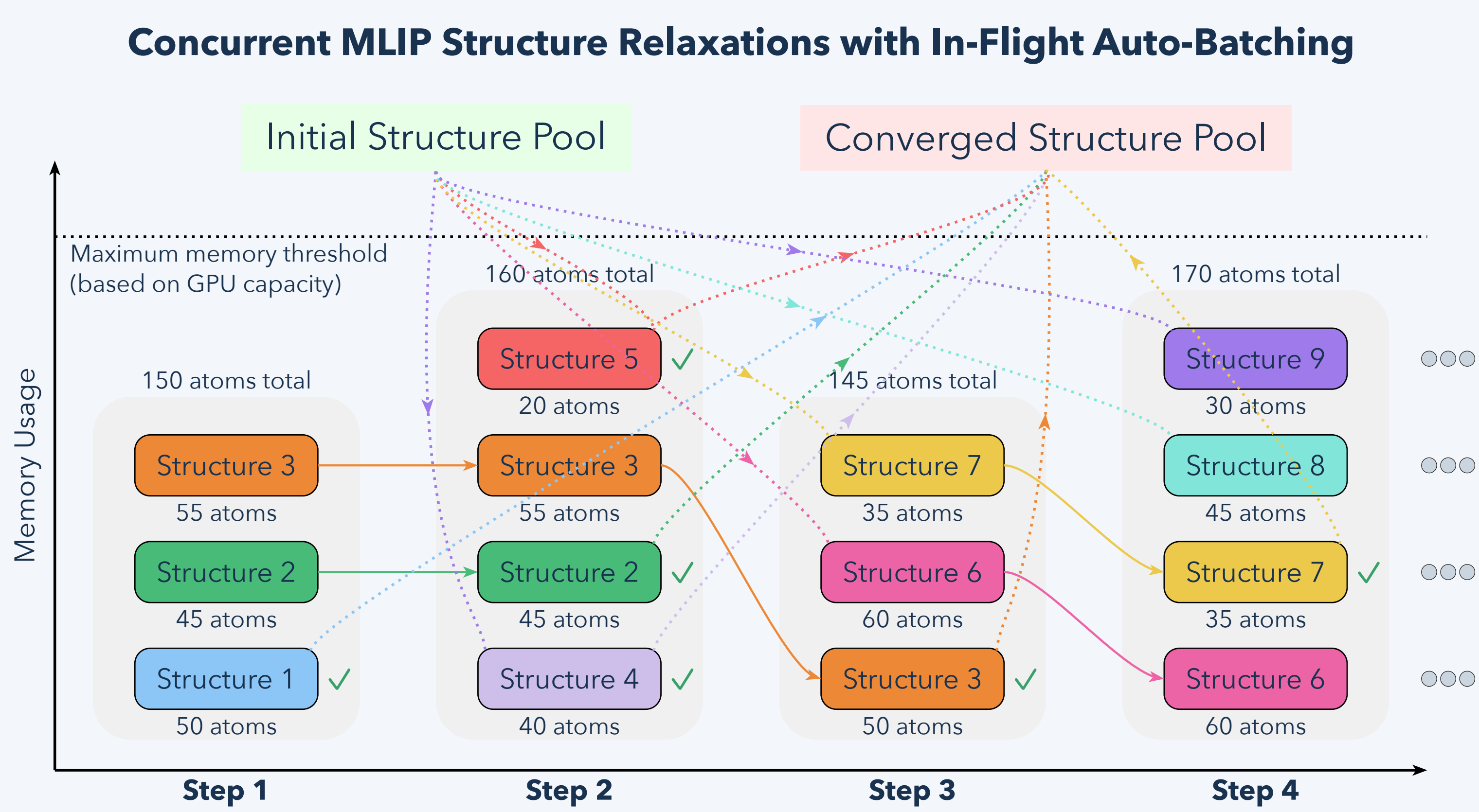
Small atomistic jobs can leave a GPU underused. Batching runs several together; replacing completed jobs keeps useful work in the batch.

1 Compare scheduling strategies



Sequential jobs leave capacity unused. Fixed batches wait for their slowest member. In-flight scheduling admits new work as individual structures finish.

2 Follow a replacement



Each structure has its own convergence state. After a structure finishes, move it to the completed pool and fill the available capacity from the waiting pool.

Batch by resource use, not just job count. Atom counts, neighbor counts, memory, and model cost constrain admission. Utilization percentages in the schematic are illustrative, not measured benchmarks.